Chandrasekar Vasan

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EDUCATION

Master of Science in Computer Science, The University of Texas at Arlington | Arlington, TX, USA **Jan 2025 - Dec 2026**
Coursework: Artificial Intelligence | Software Testing | Design & Analysis of Algorithms ***(GPA: 3.9/4.00)***
Bachelor of Technology in Information Technology, Anna University | Chennai, TN, India **Aug 2020 - Jun 2024**
Coursework: Software Engineering | Operating Systems | Cloud Computing | Machine Learning ***(GPA: 4.00/4.00)***

WORK EXPERIENCE

The University of Texas at Arlington **Aug 2025 - Present**
Graduate Research Assistant – System Administrator **Arlington, Texas, USA**

- Administered laboratory infrastructure of 60+ Linux hosts (Ubuntu, Rocky Linux 9), plus Windows systems and Jamf-managed Macs, supporting 200+ users, an Information Security Lab, and research workloads
- Automated configuration and user provisioning across 70 workstations using Ansible playbooks validated through GitLab CI before rollout, with Bash and Python scripts for Linux patching and VirtualBox VM management
- Hardened systems with SSH key-only access, firewall rules, fail2ban, and least-privilege sudo, and built Rocky Linux 9 and VirtualBox environments with managed snapshots for repeatable lab setups
- Deployed Prometheus and node_exporter monitoring across 100+ machines, routing Alertmanager alerts to PagerDuty and tuning CPU thresholds to sustained usage, which cut non-actionable notifications
- Investigated Linux and VirtualBox failures through log analysis and configuration checks, validated corrective changes, and documented diagnostic findings, recovery procedures, and operational runbooks

eNoah iSolution **Jan 2024 - Apr 2024**
Software Engineer Intern **Chennai, Tamil Nadu, India**

- Automated loan processing via React.js components and CRM APIs on PostgreSQL/MongoDB, achieving ~25% efficiency gains
- Built a loan approval and risk model comparing Logistic Regression and XGBoost, delivering ~12% higher accuracy
- Validated loan workflows across CRM and database integrations, catching rendering defects before release and documenting fixes

Kun Auto Co Pvt Ltd **Jan 2023 - Sep 2023**
Software Engineer Intern **Chennai, Tamil Nadu, India**

- Delivered a Node.js/React.js/PostgreSQL price estimation app supporting ~50 daily test scenarios for internal validation
- Tuned a Python regression model using NumPy/Pandas and hyperparameter optimization, reducing MAE by ~15%
- Set up GitHub Actions CI/CD to automatically test and deploy the application, with API endpoints and server-side validation reducing error rates

PROJECTS

CloudCache – Distributed Key-Value Store

- Built a distributed key-value store handling ~5K operations per second across replicated nodes
- Implemented Raft-based leader election and replication to maintain data consistency during node failures
- Containerized services with Docker and used Prometheus to monitor latency, replication delay, and node health

LogTrace – Centralized Log Monitoring System

- Built a centralized logging service aggregating and indexing ~10K events per minute from multiple microservices
- Implemented Elasticsearch full-text search and a React dashboard to visualize error trends
- Configured TTL retention policies, reducing stored log volume by ~25%

QueryForge – SQL Query Parser and Executor

- Built a SQL engine for CSV data, matching SQLite results on ~92% of tested SELECT, WHERE, and JOIN queries
- Developed a SQL parser with Python's PLY and added cost-based optimization to improve scan efficiency
- Added in-memory indexing, reducing query execution time by ~35% on large datasets

TECHNICAL SKILLS

- **Infrastructure & Systems:** Linux (Ubuntu, Rocky Linux), Windows, macOS (Jamf), VirtualBox, VMware, Docker, DNS, DHCP, NAT
- **Automation & Scripting:** Ansible, CI/CD (GitHub Actions), Bash/Shell, cron, Git
- **Monitoring & Observability:** Prometheus, node_exporter, Alertmanager, PagerDuty, Elasticsearch (ELK)
- **Languages & Development:** Python, C, C++, JavaScript, React.js, Node.js, Django, Flask, PostgreSQL, MongoDB, MySQL
- **Core Competencies:** System Administration, Virtualization, Troubleshooting & Root Cause Analysis, Technical Documentation

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** AWS Certified Cloud Practitioner, Google IT Support Professional, IBM Linux Command and Shell Scripting, Google Data Analytics, Certified Kubernetes Administrator (in progress)
- **Achievement:** Second place, TECHNOPHILIA 2022 symposium, for an SVM-based Parkinson's detection model (87% accuracy)